

Testing Procedure

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Definitions:

Snake: A line of x number of pixels that travels around the screen in the direction the player indicates

Goal: A single pixel showing the target for the player

Goal Area: A 1-pixel wide border around the goal where the goal is considered reached

User Buttons: The four buttons used to control the direction of the snake

Screen Edge: The boundary of the screen

Fail Screen: The display shows stripes across the screen

Procedure:

Action	Success Criteria	What it tests
Power up board	A Goal and a 5 pixel long Snake appear on the screen. The Snake starts at the lower left and begins to travel up the screen.	Initialization of the snake list, display communication, state machine game active updates
Click the right User Button	The snake turns right, first creating a right angle with its body and then becoming a line again traveling to the right.	Right User Button hardware and interrupt, Snake direction change logic

Repeat with Down, Left and Up User Buttons	The snake turns down then left than up	Down, Left and Up User Button hardware and interrupt, Snake direction change logic
While snake is going up, Click Down User Button	The snake continues to move up	Snake direction change logic
Click Right User Button, then Click Left User Button	The snake moves right	Snake direction change logic
Using the User Buttons direct the snake into each of the four Screen Edges	The screen flashes with Fail Screen (stripes). Then the game starts over with a five-pixel snake traveling up from the lower left corner	Game fail logic, end_game() function and reset_game() function
Using the User Buttons direct the snake towards the Goal	When the Goal Area is reached the goal disappears and the over the next three updates the snake grows by three pixels. A new Goal appears somewhere else on the screen and the snake continues to move.	process_goal(), within_goal_range(), generate_goal()
Repeat the previous step for 2 more goals	The snake should become 6 pixels longer and new goals should always be on the screen and not at the outermost row of pixels on any screen edge.	process_goal(), within_goal_range(), generate_goal()
Using the User Buttons, crash the snake into an edge	The screen flashes with Fail Screen. Then the game starts over with a five-pixel snake traveling up from the lower left corner	Snake length reset. Game fail logic, end_game() function and reset_game() function.